

Week 18 Report - Internal Testing, Parameter Optimisation & Pilot Usability (VerifAI)

Overview

In Week 18, internal testing, parameter optimisation, and a pilot usability study were conducted to evaluate the robustness, configurability, and usability of the VerifAI deepfake detection framework.

Internal Testing

The FastAPI backend was tested using a dedicated internal test script and a health-check endpoint. Core functionality was verified using valid MP4 inputs, while multiple edge cases were evaluated, including invalid file types, malformed MP4 files, and empty uploads. API-level validation was added to ensure invalid inputs return appropriate HTTP 400 errors instead of internal server failures, improving system robustness.

Parameter Optimisation

A small, controlled tuning pass was performed, focusing on inference-time parameters. The number of sampled video frames ($n_frames = 3, 5, 7$) was evaluated using consistent test videos to isolate parameter effects. Results showed stable predictions across configurations, with small measurable probability variations on a second test video. Based on these experiments, $n_frames = 5$ was retained as a balanced default. Decision threshold behaviour was also evaluated to support interpretability.

Pilot Usability Study

A pilot usability study was conducted with five users (two technical and three non-technical). All participants completed the task of running a prediction via the Swagger UI. Feedback highlighted that while technical users navigated the interface easily, non-technical users found the JSON output and probability interpretation unclear. These findings informed documented improvements and plans, including clearer output explanations and a user-friendly frontend.

Outcome

Week 18 successfully demonstrated API reliability, controlled parameter tuning with justification, and actionable usability insights, providing a strong foundation for further refinement of VerifAI.